

A First-Lower-Sideband Orbit Atom and a Mandatory Compensation Obstruction

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Abstract

The one-alias off-background of RH-330 contains every coefficient below the critical order $2k$ except the critical coefficient itself. We prove that the first lower even sideband $n_- = 2k - 2$ already contains an alias-scale physical boundary-orbit atom. The primitive period- $2(k - 1)$ boundary orbit has a folded subset of $2k - 3$ marked points in the frozen RH-334 far set. Its noisy localized trace is exactly zero, while its deterministic mass D_{k-1}^{orb} satisfies

$$D_{k-1}^{\text{orb}}/H_{k-1} \longrightarrow +\infty, \quad H_{k-1} = (k-1)R^{-2(k-1)}.$$

Writing the actual typed sideband coefficient as $q_- = -D_{k-1}^{\text{orb}} + C_-$, vanishing of the off-alias weighted prefix necessarily forces $C_- = D_{k-1}^{\text{orb}} + o(H_{k-1})$, at relative precision $o((\beta R)^{-2(k-1)})$. Thus any proof that takes separate absolute values after the orbit/complement split fails at this mandatory sideband. This is not a positive lower bound for the fully signed coefficient: the complement may cancel the atom. The repository provides no moving-order estimate for that complement, so both off-alias vanishing and off-alias nonvanishing remain not testable. The physical counterloop radial sideband is retained without a sign claim because the printed value $C_M > 1$ lacks an interval certificate. No determinant or Riemann-hypothesis conclusion is obtained.

1 Typed off-alias coefficient

Work on the physical natural clock

$$k = k_\sigma \in \mathbb{N}, \quad k \geq 2, \quad k = \frac{\log(1/\sigma)}{2 \log \lambda} + O(1), \quad \sigma \rightarrow 0, \quad H_k = kR^{-2k}, \quad R = \frac{7}{5}. \quad (1)$$

Choose any integer cut

$$2k < h_\sigma \leq 4k. \quad (2)$$

For the actual Hardy full-trace constituent $q_{\sigma,k,n}$, RH-330 defines the nonnegative off-alias prefix

$$\mathcal{E}_{\sigma,\text{off}}^{(h)}(R) = \sum_{\substack{2 \leq n < h_\sigma \\ n \neq 2k}} \frac{|q_{\sigma,k,n}| R^n}{n} \quad (3)$$

[7].

We retain the corrected RH-334 folded backward-observable operator K_σ , the frozen basepoint partition $[0, 1] = J^- \sqcup J^+ \sqcup F$, and the localized raw slots $\mathcal{B}_{\sigma,k,n}, \mathcal{S}_{\sigma,k,n}, \mathcal{R}_{\sigma,k,n}$ [5]. For every $n \geq 2$, define

$$q_{\sigma,k,n} = c_{\sigma,n}^H - s_{k,n} - a_n^{\text{num}}, \quad (4)$$

$$\mathcal{P}_{\sigma,n} = r_H^{-n} \{(-1)^n - \lambda_-(\sigma)^n\}, \quad r_H = \frac{17}{20}, \quad (5)$$

$$\mathcal{A}_{k,n} = s_{k,n} - p_n^{\text{pole}}. \quad (6)$$

Proposition 1.1 (All-order typed five-slot identity). *For every $n \geq 2$,*

$$q_{\sigma,k,n} = \mathcal{B}_{\sigma,k,n} + \mathcal{S}_{\sigma,k,n} + \mathcal{R}_{\sigma,k,n} + \mathcal{P}_{\sigma,n} - \mathcal{A}_{k,n}. \quad (7)$$

The coefficient type is the Hardy full-trace constituent, not automatically the modulus-complement coefficient.

Proof. The deterministic numerator anchor is $a_n^{\text{num}} = c_n^H - p_n^{\text{pole}}$. Hence

$$q_{\sigma,k,n} = (c_{\sigma,n}^H - c_n^H) - (s_{k,n} - p_n^{\text{pole}}).$$

The exact bulk split gives

$$c_{\sigma,n}^H - c_n^H = r_H^{-n} \{\text{Tr } K_\sigma^n - P_n\} + \mathcal{P}_{\sigma,n}.$$

RH-334 partitions the raw trace into $\mathcal{B} + \mathcal{S} + \mathcal{R}$ for every $n \geq 2$. Substitution proves (7). Identifying this constituent with the modulus complement would additionally require the noisy-head defect to vanish, which is not assumed here. $\square$

Set

$$n_- = 2k - 2 = 2m, \quad m = k - 1, \quad H_m = mR^{-2m}. \quad (8)$$

For every admissible cut (2), $2 \leq n_- < h_\sigma$ and $n_- \neq 2k$. Therefore the corresponding summand is always present in (3).

At this sideband the exact RH-326 counterloop identity is

$$\mathcal{A}_{k,n_-} = 2(\beta^{n_-} - \beta_k^{n_-}), \quad \beta_k = \frac{|M_k|^{-1/(2k)}}{r_H}, \quad \beta = \frac{1}{r_H \sqrt{\lambda}} \quad (9)$$

[4]. RH-17 proves only $C_M > 0$ analytically in $|M_k| = C_M \lambda^k (1 + o(1))$. Its printed value 1.9463429052... is not an interval certificate, so no eventual sign of (9) is used. Here “alias-scale” refers only to the size of the orbit atom. The lower sideband is not an alias impulse of the current period- $2k$ counterloop; (9) is its radial counterloop contribution.

2 A physical orbit atom at $n_- = 2k - 2$

Let p_{2m} be the primitive boundary point of physical period $2m$ and write $x_j = f^j(p_{2m})$. RH-17 supplies the ordered component chain, fixed gaps from the fold cusp b , and the multiplier law

$$M_m = (f^{2m})'(p_{2m}) = -C_M \lambda^m \{1 + o(1)\} \quad (10)$$

[1]. Define

$$\Omega_m = \{|x_j| : 0 \leq j < 2m, j \neq 2m - 2\}. \quad (11)$$

Theorem 2.1 (Mandatory lower-sideband far atom). *For every fixed RH-334 window parameter $A > 0$ and all sufficiently large k ,*

$$\Omega_m \subset F_{\sigma,k,A}, \quad |\Omega_m| = 2m - 1 = 2k - 3. \quad (12)$$

Define

$$D_m^{\text{orb}} = r_H^{-2m} \frac{2m - 1}{1 + |M_m|} > 0. \quad (13)$$

Then the localized sideband far subledger is exactly

$$\mathcal{R}_{k,-}^{\text{orb}} := r_H^{-2m} \{L_{\sigma,2m}(\Omega_m) - P_{2m}^{\text{abs}}(\Omega_m)\} = -D_m^{\text{orb}}, \quad (14)$$

and

$$\frac{D_m^{\text{orb}}}{H_m} = \frac{2}{C_M} (\beta R)^{2m} \{1 + o(1)\} \longrightarrow +\infty. \quad (15)$$

Proof. The fixed-gap proof of RH-338 applies to the period- $2m$ orbit even though the windows are evaluated on the narrower k -clock [3]. The endpoint lies to the right of b by a fixed gap; all retained internal even points are at most $h(b) < b$; and all odd folded points are at most $r < b$. Since $A\sqrt{\sigma} \rightarrow 0$, deleting only $h(p_{2m})$ leaves $2m - 1$ distinct points in $F_{\sigma,k,A}$.

The finite set Ω_m has zero multiplication operator on L^2 , so its noisy localized trace is zero. Folding preserves the multiplier and marked multiplicity. Since $M_m < 0$ for large m , each deterministic weight is $1/(1 + |M_m|)$, proving (14). Finally (10) and $\beta^{2m} = r_H^{-2m} \lambda^{-m}$ give

$$D_m^{\text{orb}} = \frac{2m}{C_M} \beta^{2m} \{1 + o(1)\}.$$

Division by H_m gives (15); RH-336 proves $\beta R > 1$ exactly [6]. $\square$

As in RH-338, Ω_m is only an analytic subpartition of the already frozen far set. It is not a new canonical physical observation window.

3 Off-alias compensation obstruction

Collect every other signed contribution at n_- into the exact complement

$$C_k^- := q_{\sigma,k,n_-} + D_m^{\text{orb}}. \quad (16)$$

Thus

$$q_{\sigma,k,n_-} = -D_m^{\text{orb}} + C_k^-. \quad (17)$$

C_k^- includes the rest of the raw far slot, the boundary and sibling slots, the parity packet, and the radial counterloop term (9), with their actual signs retained.

Theorem 3.1 (Necessary compensation for off-alias vanishing). *For every admissible cut (2),*

$$\mathcal{E}_{\sigma,\text{off}}^{(h)}(R) \geq \frac{|q_{\sigma,k,n_-}| R^{2m}}{2m} = \frac{|q_{\sigma,k,n_-}|}{2H_m}. \quad (18)$$

Consequently,

$$\mathcal{E}_{\sigma,\text{off}}^{(h)}(R) \rightarrow 0 \implies C_k^- = D_m^{\text{orb}} + o(H_m). \quad (19)$$

The required relative precision is

$$o\left(\frac{H_m}{D_m^{\text{orb}}}\right) = o((\beta R)^{-2m}). \quad (20)$$

Moreover, an atom/complement proof that takes separate absolute values before their signed sum cannot close, because the orbit atom alone contributes

$$\frac{D_m^{\text{orb}} R^{2m}}{2m} = \frac{D_m^{\text{orb}}}{2H_m} \rightarrow +\infty. \quad (21)$$

Proof. The n_- summand is present in (3), proving (18). If the nonnegative prefix tends to zero, that summand tends to zero and hence $q_{\sigma,k,n_-} = o(H_m)$. Substitute (17) to obtain (19). The reciprocal of (15) gives (20); (21) is the same identity without the signed complement. $\square$

This theorem is one-way. It does not lower-bound the fully signed $|q_{\sigma,k,n_-}|$, because C_k^- may compensate the orbit atom. RH-19 leaves the requisite bulk moving-order control open [2]. Therefore neither $\mathcal{E}_{\sigma,\text{off}}^{(h)}(R) \rightarrow 0$ nor its failure follows from the current repository. The exact verdict is a scoped obstruction to the separate-absolute sideband route and **NOT_TESTABLE** for the aggregate off-alias background.

4 Reproduction and claim boundary

The artifact evaluates the physical algebraic constants and period- $2m$ boundary orbits at high Decimal precision for $k = 3, 5, 9, 17, 33$ and $A = 1/4$. It reproduces the mandatory sideband orders, far counts 3, 7, 15, 31, 63, orbit closure, the identity $R^{2m}/(2m) = 1/(2H_m)$, and the super-target orbit mass. These finite rows are diagnostic formula checks, not interval certificates or moving-order evidence.

The paper proves only the actual typed lower-sideband atom and the necessary signed compensation condition. It proves neither off-alias vanishing nor a positive lower bound for the fully signed sideband coefficient. It does not close the critical packet, noisy head, counterloop synchronization, determinant gluing, full-trace replacement, or full-trace divergence. No Hilbert–Polya operator, Riemann-zero identification, von Mangoldt trace, completed-zeta divisor equality, or Riemann-hypothesis proof is obtained. Gates A–E remain false/open.

References

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